AARAV SHAH

(951) 906-2964 | ashah264@ucr.edu | Riverside, California

EDUCATION

University of California, Riverside
B.S. in Computer Science | Expected 2029
Relevant Coursework: Data Structures, Algorithms, Statistical Analysis, Machine Learning

RESEARCHEXPERIENCE

Independent Researcher | Inventory Forecasting Using ARIMA Models (2024)
• Conducted quantitative research on time-series forecasting using ARIMA statistical models for inventory optimization
• Authored technical research paper analyzing demand forecasting techniques, reducing prediction error by 25% compared to baseline models
• Performed comprehensive data analysis and statistical modeling using Python (Pandas, Matplotlib, Statsmodels)
• Developed data visualization dashboards to communicate complex statistical findings to non-technical stakeholders
• Applied rigorous experimental methodology including hypothesis testing, model validation, and sensitivity analysis

AWS | National University of Singapore | AI & Machine Learning Research Specialist (Dec 2023)
• Led research initiatives in predictive analytics achieving 99.8% model accuracy, outperforming baseline by 15%
• Designed and implemented ML research pipeline using Random Forest, KNN, and Neural Networks, reducing computational time by 40%
• Conducted large-scale data preprocessing and analysis, increasing dataset processing efficiency by 60%
• Developed NLP-based text classification models achieving 95% precision in sentiment analysis applications
• Collaborated with cross-functional research teams to solve complex computational problems using AWS infrastructure

Computer Vision Research | Code Day Hackathon (2022) — 1st Place Winner
• Designed and implemented computer vision algorithms for real-time gesture recognition with 92% accuracy
• Optimized image processing pipeline achieving 30 FPS performance on standard hardware
• Conducted rapid prototyping and iterative testing in collaborative 24-hour research sprint
• Produced comprehensive technical documentation and presentation of research findings

TECHNICALSKILLS

Programming Languages: Python, JavaScript, SQL, Java, Node.js, Express.js, Flask, Restful API's
Data Analysis & Research Tools: Pandas, NumPy, Matplotlib, Statsmodels, Scikit-learn, TensorFlow
Statistical Methods: ARIMA modeling, regression analysis, hypothesis testing, time-series analysis, predictive modeling
MachineLearning:RandomForest,KNN,NeuralNetworks,NLP,sentimentanalysis,computervision
Research Tools: Jupyter Notebooks, Git/GitHub, AWS (SageMaker, EC2), data visualization
Databases: SQL, MangoDB, NoSQL, data pipeline optimization
Additional: Experimental design, technical writing, literature review, quantitative analysis

RELEVANTTECHNICALEXPERIENCE

Full Stack Development Certification | IBM (2024)
• Developed 5+ data-driven web applications with responsive interfaces and optimized backend systems
• Built RESTful APIs serving 10K+ daily requests with optimized database queries (45% response time improvement)
• Implemented CI/CD pipelines and version control best practices for reproducible research workflows
• Deployed cloud-based solutions on IBM Cloud and AWS with 99.9% uptime reliability

Assistive Technology Research | Product Developer (2023)
• Designed and tested accessibility software using nose tracking and voice recognition algorithms
• Conducted user research and iterative testing with 15+ participants achieving 88% task completion rate
• Applied human-computer interaction principles reducing user learning curve by 70%
• Performed quantitative usability analysis demonstrating product-market validation

LEADERSHIP&COLLABORATION

Curriculum Manager | Daricha Organization (Apr 2024 - Present)
• Manage remote education program serving 200+ students with data-driven curriculum improvements
• Coordinate interdisciplinary team of 10+ educators across multiple time zones
• Increased course completion rate by 85% through systematic analysis and optimization

Initiative Leader | USIC Rotary Club (2022 - 2024)
• Led 8+ community research and service initiatives impacting 500+ families
• Organized data collection and impact assessment campaigns increasing participation by 60%

PUBLICATIONS&PRESENTATIONS

Technical Paper: "Inventory Forecasting Using ARIMA: A Quantitative Analysis of Demand Prediction Techniques"(2024)

Silver Medalist, 5th International STEM Olympiad (Coding Category), Frankfurt, Germany – 2025